

12 May 2026

Supplemental Materials For Torres Honor Thesis

1. Introduction

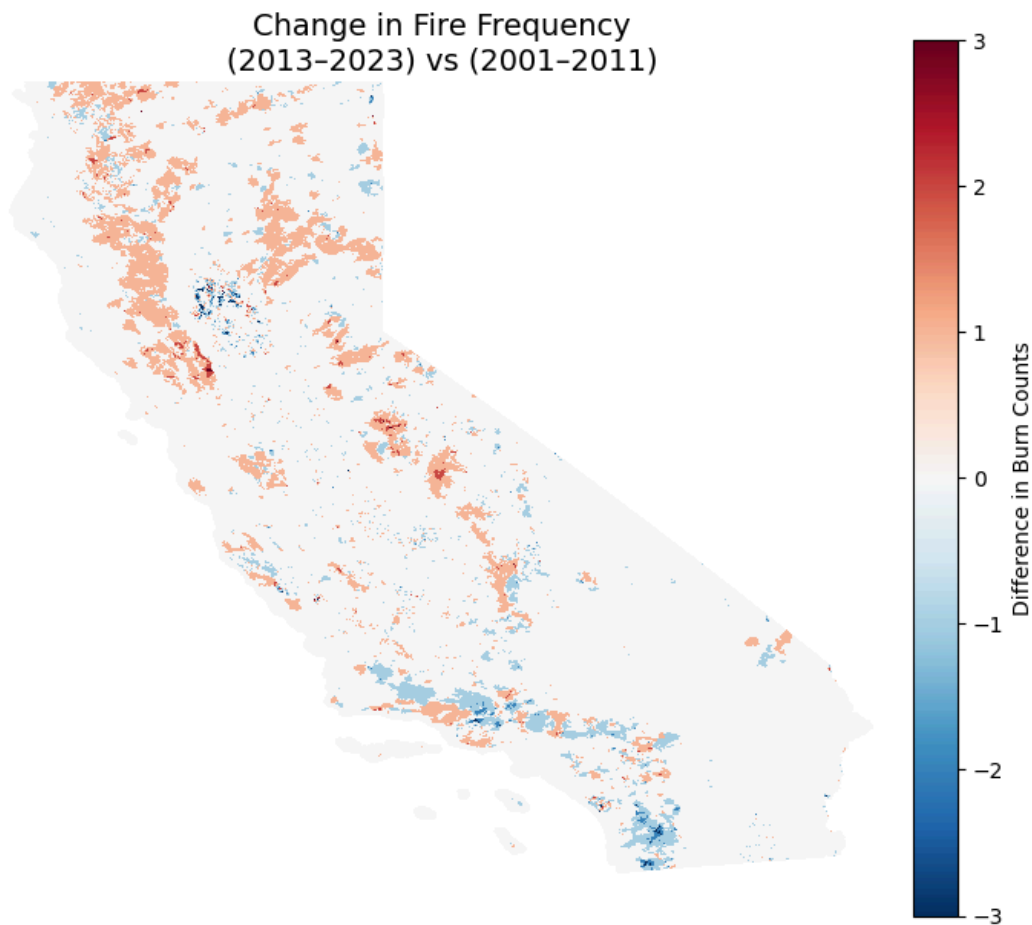


Figure 1: Change in fire frequency across California comparing the early (2001–2011) and late (2013–2023) periods of the study window. Red indicates pixels with more frequent burning in the later period, blue indicates fewer burns. The general shift toward increased fire frequency in the northern Sierra Nevada and Cascade ranges is consistent with documented trends in western US fire regimes.

2. Data and Methods

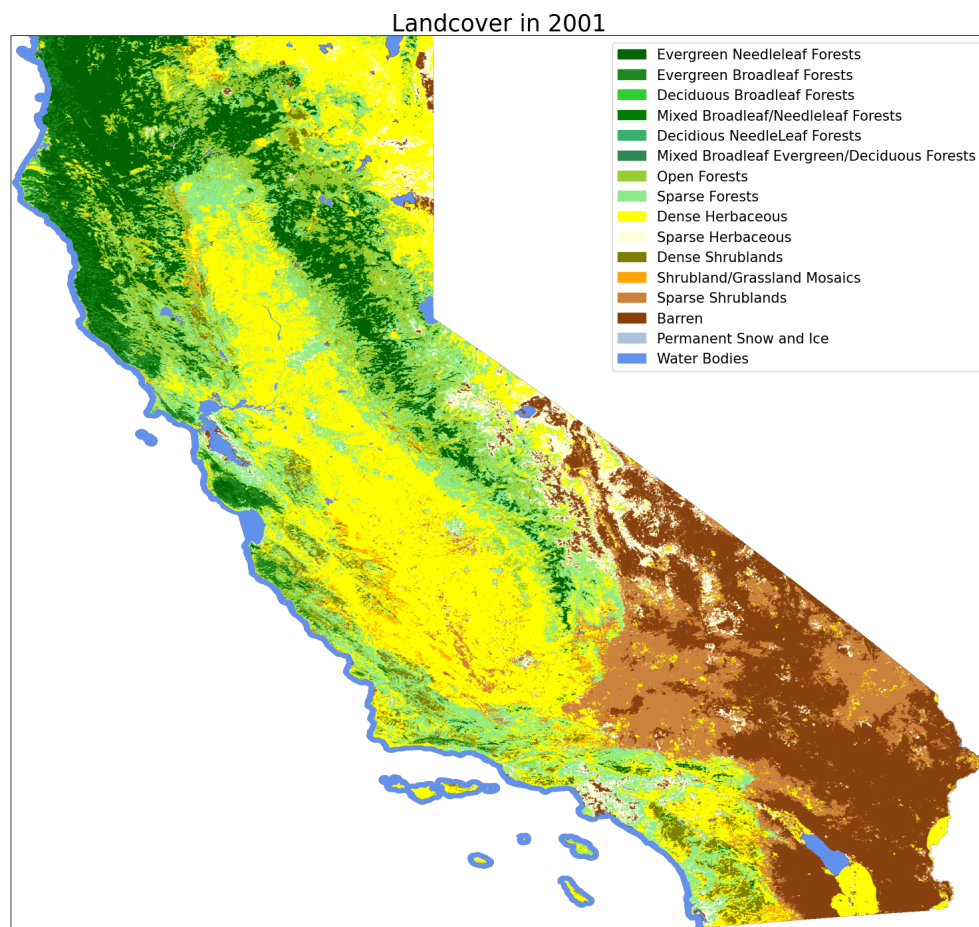


Figure 2: Landcover classifications from the MODIS Landcover product visualized across the state for the year 2001.

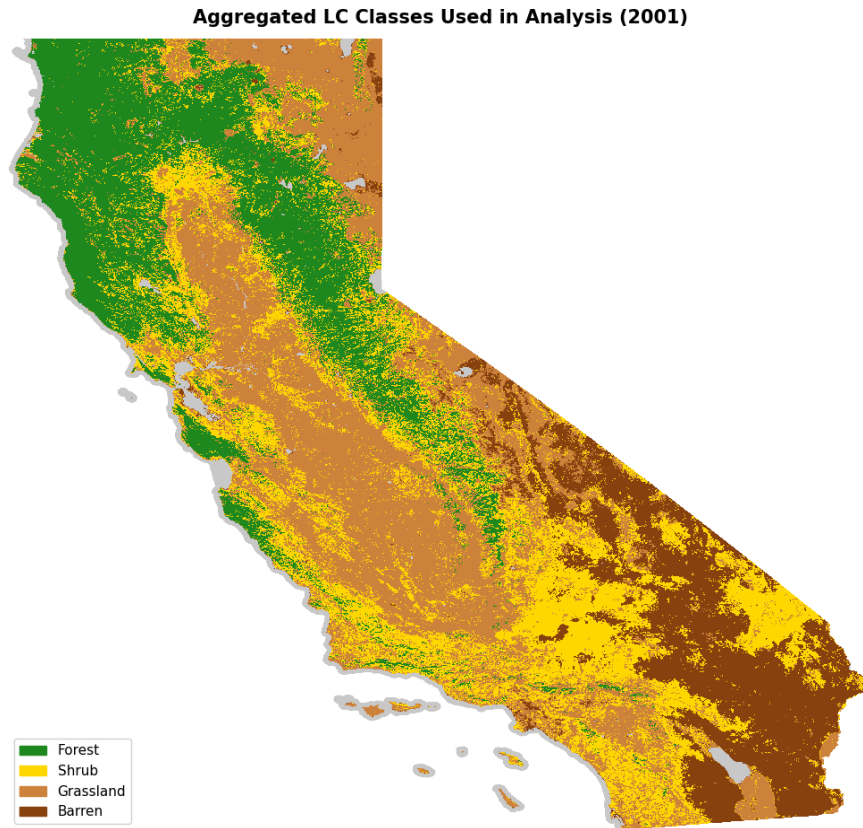


Figure 3: An aggregated version of original landcover classifications used for analysis of landcover succession pathways and for identified burned forest areas for recovery modeling, visualized for 2001.

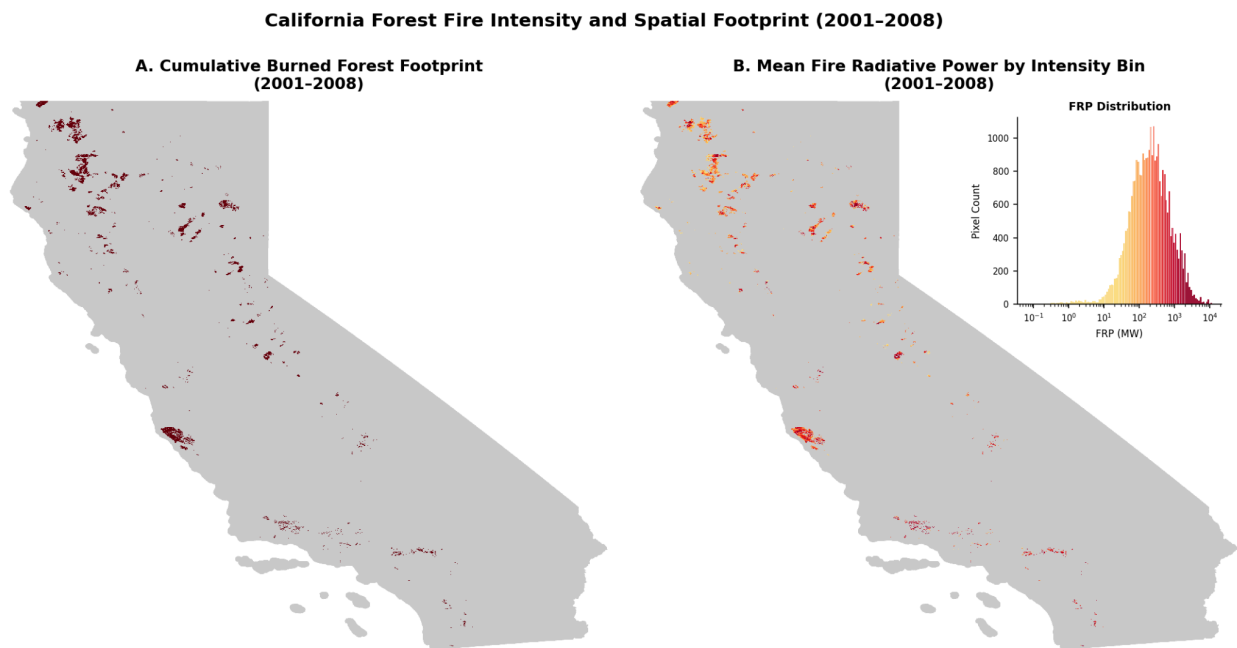


Figure 4: Spatial distribution of burned areas and mean fire radiative power for 2001-2008. These pixels were then filtered to be further constrained to have 15 years of undisturbed recovery, along with other criteria.

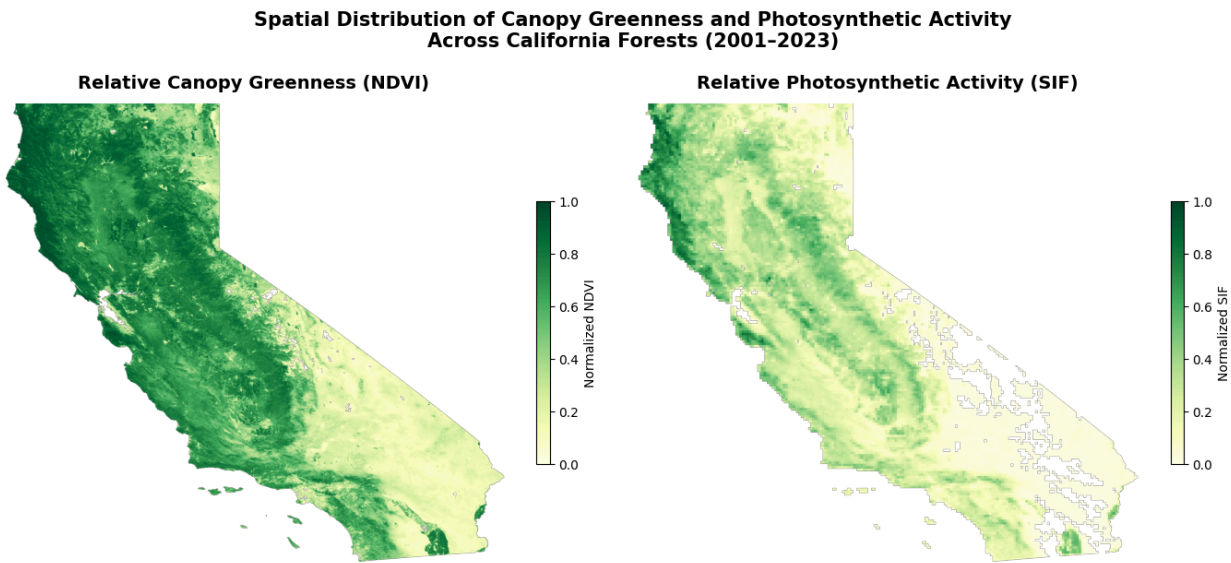


Figure 5: Relative NDVI and SIF across California forest pixels (2001–2023), illustrating the spatial patterns of the two metrics used in this study. No-data values in the SIF panel reflect areas of sparse or absent vegetation.

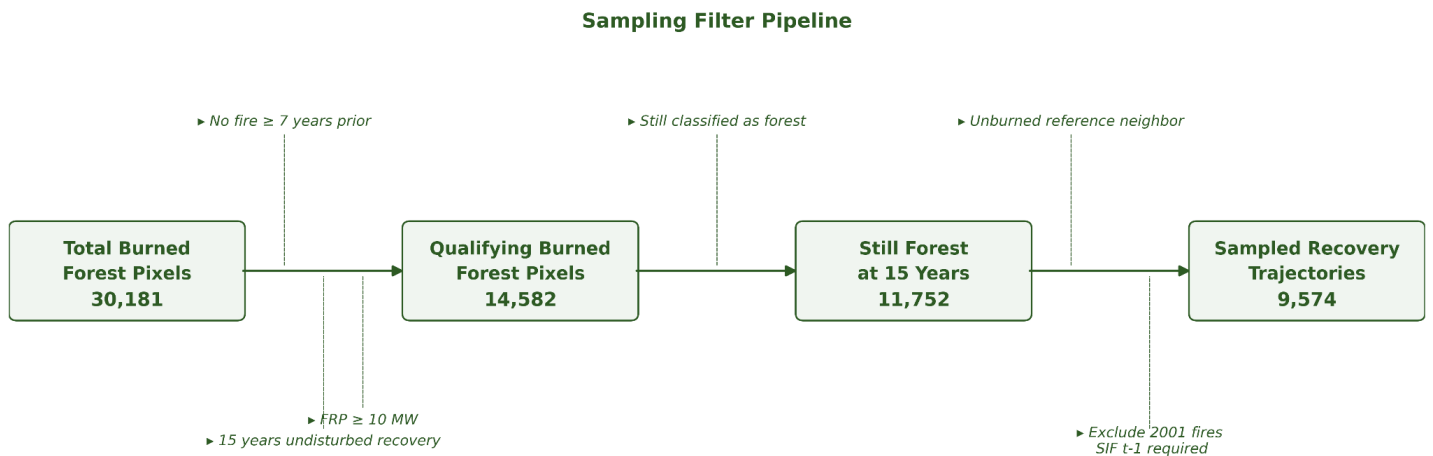


Figure 6: Full sampling pipeline for various stages of criteria.

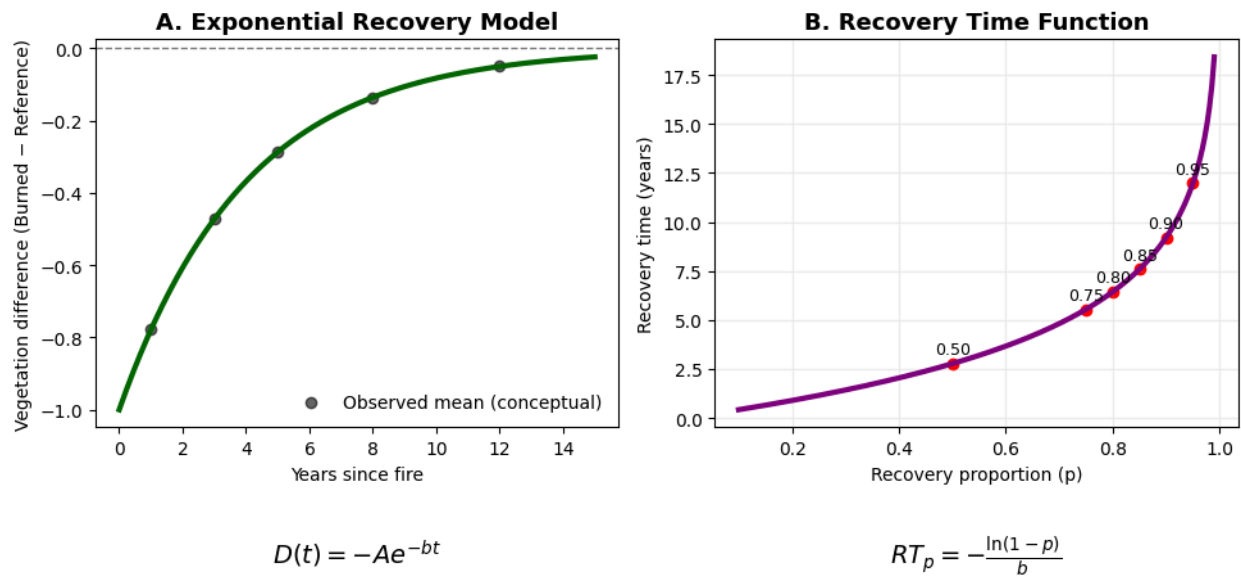


Figure 7: a) Negative exponential growth model fitted to conceptual observed mean values, and b) recovery time calculation to various thresholds of an unburned reference, obtained by solving for RT_p .

3. Results

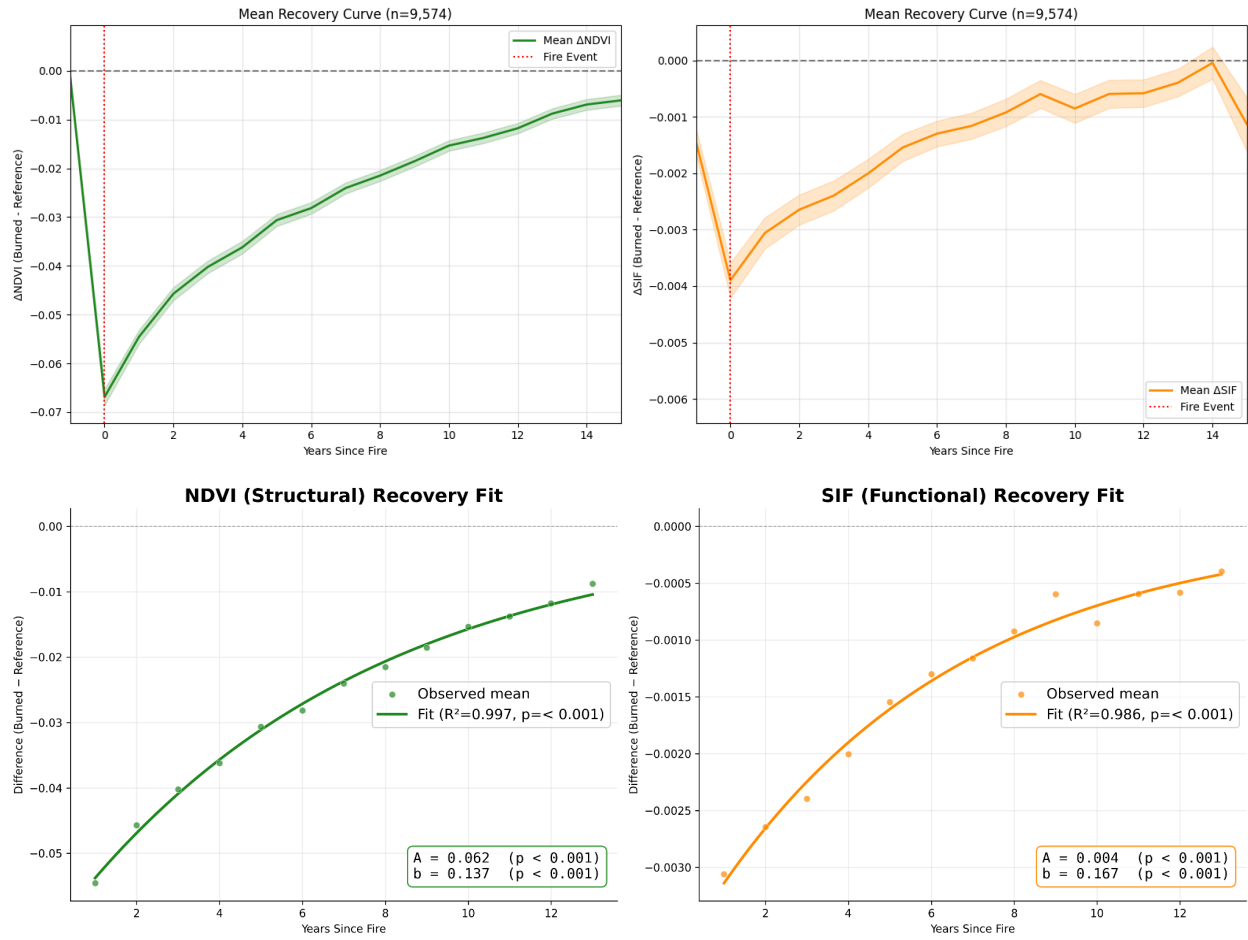


Figure 8: Mean post-fire recovery trajectories for NDVI (left) and SIF (right) across 9,574 burned forest pixels in California (2002–2023). Observed mean values (points) and fitted negative exponential model (line) are shown for each metric. Shaded regions represent 95% confidence intervals. Trajectories are expressed as the difference between burned pixels and unburned reference pixels, with zero indicating full recovery to pre-fire reference levels.

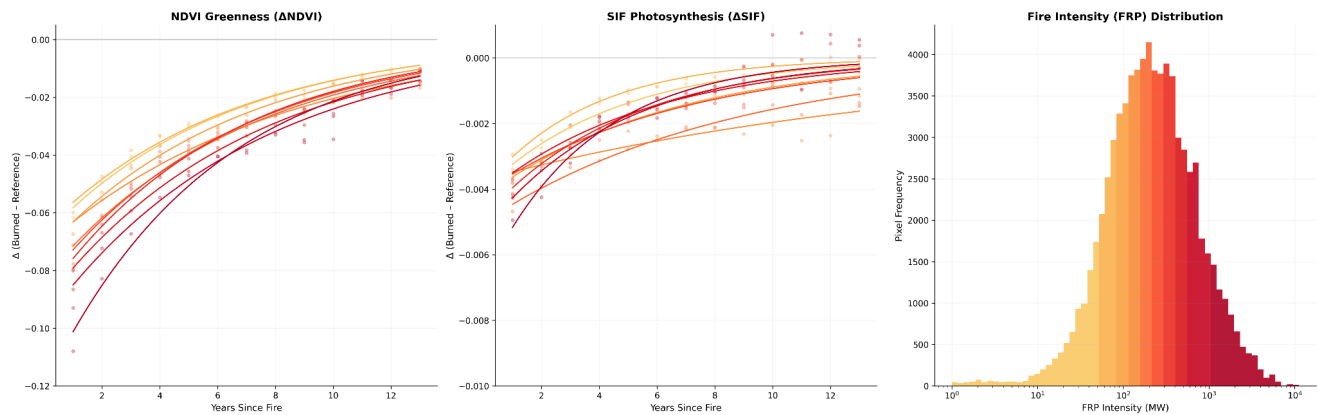


Figure 9: Mean recovery trajectories for equal sample, log-based 10 intensity classes.

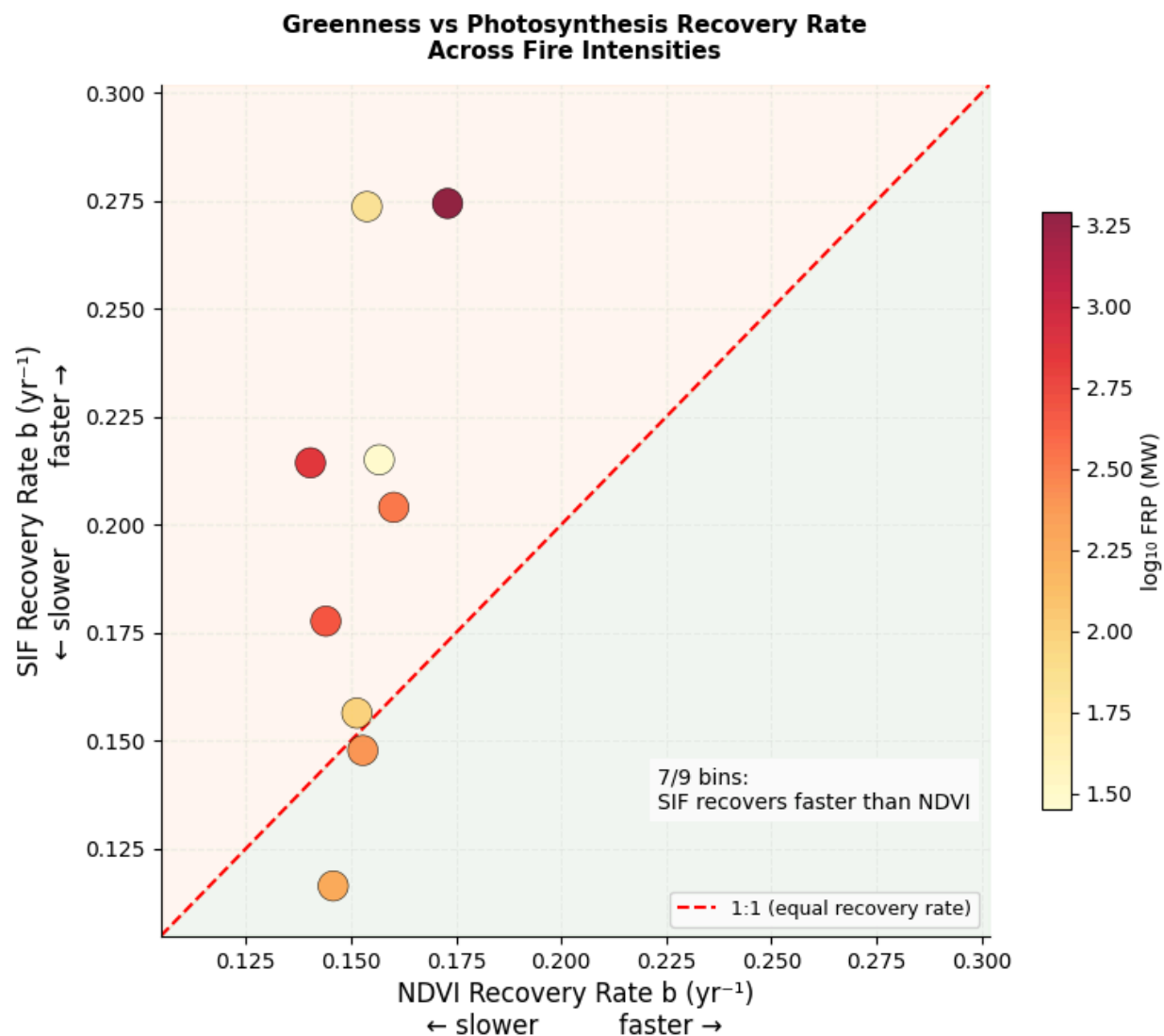


Figure 10: SIF vs. NDVI exponential recovery rates (b , yr^{-1}) across fire intensity bins. Each point represents fires binned by $\log_{10}(\text{FRP})$, colored by fire intensity. Points above the 1:1 line indicate faster functional (SIF) than structural (NDVI) recovery. In 7 of 9 good fits, SIF recovery rates exceeded NDVI, suggesting photosynthetic capacity is re-established before canopy greenness following fire.